



Glenohumeral Arthritis in the Athlete

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Arthritis or degenerative joint disease (DJD) of the shoulder in young athletes is a disabling condition that often limits sports participation and is extremely challenging to treat. Shoulder arthritis, defined as a degeneration of the articular cartilage of either the humeral head or glenoid, is caused by multiple etiologies, including primary osteoarthritis and secondary causes such as trauma, iatrogenic disease, and rheumatoid disease. Although each subtype has a unique set of considerations and treatment options, all have the same end goal of providing a functional, pain-free shoulder that has the durability to withstand the forces of daily and recreational activities for the duration of the athlete's life—a particularly important and challenging consideration in this young, active, population.

Shoulder DJD begins with loss of mobility and the inability to maintain function because of contracture and pain in and around the shoulder. This condition eventually leads to difficulty in continuing recreational activities, which is a substantial hurdle for individuals who wish to lead an active lifestyle. A wide spectrum of disease can be present, from focal cartilaginous softening to diffuse, full-thickness lesions of both the glenoid and humeral head.

EPIDEMIOLOGY

In the United States the overall incidence of glenohumeral chondral lesions and their natural history are largely unknown. These lesions are often found incidentally at the time of arthroscopy for other pathologies. The causes of these lesions range from idiopathic to specific underlying pathologies that eventually lead to osteoarthritis. Chondrolysis, a specific form of osteoarthritis, has been studied in national registries and has an incidence of 5.5 per 10 million person-years. It has an increased incidence in persons with diabetes and in those who have undergone an orthopedic procedure.¹ Nakagawa et al.² studied primary osteoarthritis, specifically excluding patients with rheumatoid disease and those with rotator cuff arthropathy, and found an incidence of 0.4% in persons with orthopedic problems versus 4.6% in those with shoulder disease presentation.

Primary osteoarthritis is typically found in persons older than 60 years.²⁻⁴ In the young, healthy, and active population, the incidence is expected to be much lower, although specific data are lacking in the current literature. Cadaveric studies have shown an increased incidence of chondral lesions associated with rotator cuff tears, with glenoid defects in 32% and humeral defects found

in 36% compared with rates of 6% and 7%, respectively, in shoulders without rotator cuff tears.⁵

Posttraumatic osteoarthritis, specifically those related to shoulder dislocations, have been studied extensively to look at chondral injuries. In 1997 Taylor and Arciero⁶ reported that 57 of 63 patients had a Hill-Sachs lesion after anterior dislocation; 40% were chondral lesions, and 60% were osteochondral lesions. Furthermore, arthroscopic evaluation of 212 patients demonstrated a 23% incidence of glenoid defects and an 8% incidence of humeral head defects in persons who had one dislocation, whereas persons who reported more than two dislocations had a significantly increased incidence of arthritic lesions (27% and 36%, respectively).⁷

CLASSIFICATION

Primary Osteoarthritis

Primary shoulder osteoarthritis is a rare phenomenon in the young athlete; this disease occurs mostly in patients 60 years and older, and its etiology is unknown. It begins with glenohumeral joint stiffness, followed by joint space narrowing and humeral head and inferior glenoid osteophytes, which is considered the classic finding, with an intact rotator cuff.⁸ Samilson and Prieto⁹ first described radiographic parameters for glenohumeral arthritis in 1983. Since then, the classification has been revised, and in 2002 Guyette et al.¹⁰ further refined this classification, with radiographs evaluated for humeral osteophytes, glenoid changes, and loss of joint space, collectively.

Walch et al.¹¹ have further noted that fixed posterior humeral head subluxation may precede the development of osteoarthritic changes and may be the first sign of primary glenohumeral arthritis, and Ropars et al. indicated that a spinoglenoid cyst is an even earlier sign of glenohumeral degeneration.¹² Patients often have anterior capsular tightness that causes a static subluxation of the humeral head in relation to the glenoid. As arthritis progresses beyond the 55% humeral subluxation index, described by Kidder,¹³ further osteophyte formation develops, including progressive central loss of cartilage on both the glenoid and humeral head.

Instability Arthropathy

Although the rates of shoulder arthritis have been demonstrated to be higher in patients who have sustained a shoulder dislocation (so-called *dislocation arthropathy*) and in patients who have

Abstract

This chapter will review current concepts in management options for glenohumeral osteoarthritis of the young athlete. Glenohumeral arthritis is a complex problem in the young athlete. It has multiple etiologies and basic principles for both operative and nonoperative treatment options are changing daily with new treatment options. A systematic review of current literature was done for all nonsurgical and surgical options of glenohumeral arthritis in patients <50 years of age. Osteoarthritic options from pathologies to include dislocation arthropathy, capsuloraphy arthropathy, avascular necrosis, osteonecrosis, rheumatoid arthritis, and chondrolysis are discussed with presentation, evaluation, and surgical versus nonsurgical options covered. Early nonsurgical options with individualized arthroscopic and open surgical options for uncontrolled pain are discussed. An author's preferred treatment option and complications are further defined with a treatment algorithm provided. There are multiple treatment options that must be individually tailored to patient expectations, age, and activity level. Some patients are able to return to prior level of activity, while others are limited by implant constraints and pain modulation modalities.

Keywords

young athlete arthritis
dislocation
arthropathy
humeral head
chondromalacia
athletic shoulder pain

had that dislocation treated surgically (so-called *capsulorrhaphy arthropathy*), we believe that these entities, although deserving of discrete consideration, may have significant overlap and are various points on a spectrum of the same disease process. However, to date, the literature has separated these processes into two distinct entities.

Dislocation Arthropathy

Impaction and shear across the glenohumeral joint causes an articular cartilage lesion during shoulder dislocation, with an incidence of chondral and osteochondral lesions of 47% and 46%, respectively.^{6,14} Initially described by Neer et al. in 1982, Samilson and Prieto classified the condition in 1983.^{9,15} This form of arthritis is associated with increasing age at the initial event, direction of dislocation (with posterior causing more changes than anterior), and associated glenoid fractures.^{16–18} However, the number of dislocations and prior stabilization procedures were not associated with development of shoulder arthritis.⁹ Interestingly, Matsoukis et al.¹⁹ reported no difference between patients treated with and without surgery in regard to the severity of arthritis, and the overall incidence of arthritis after shoulder dislocation in patients treated nonoperatively ranges from 10% to 20%.^{16–18} Most recently, 25-year follow-up data, reported by Hovelius and Saeboe²⁰ in 257 shoulders that sustained primary anterior dislocation, show that shoulders were radiographically normal in only 44% of patients at 25 years and the arthrosis development was related to age at primary dislocation (>25 years), recurrence of instability, participation in high-energy sports causing injury, and an alcohol abuse history. Although the rate of moderate to severe arthritis was 18% in shoulders without a recurrence compared with 26% with surgically stabilized shoulders, the rate of arthrosis in these surgically stabilized shoulders was less than in shoulders that were reported to have “stabilized over time,” raising a significant question regarding the role of recurrent instability in the future development of arthrosis.

Capsulorrhaphy Arthropathy

Capsulorrhaphy arthropathy, as coined by Matsen and Wirth,²¹ is described as arthrosis that is attributed to overtightening of the capsular structures which results in abnormal translation of the humeral head opposite the direction of capsulorrhaphy. Found in male patients at a mean age of 45, it has been linked to length of time since follow-up, amount of external rotation contracture, and the age at initial trauma.^{16,22} Biomechanical testing of selective capsular plication has shown that this mechanism causes alterations in humeral head translation.^{23–26} These translational differences lead to nonanatomic biomechanics, asymmetric cartilage wear, and ultimately arthritis (Fig. 52.1).

Surgical options have evolved as we have come to better understand the effects of biomechanics on shoulder articulation. Surgical options can now be divided into anatomic and non-anatomic procedures. Most nonanatomic repairs have been abandoned, including the Putti-Platt²⁷ and Magnuson-Stack methods²⁸; however, the Bristow-Latarjet method^{29,30} (although modified from its original description)³¹ has continued to be a mainstay of treatment in cases of anterior instability in the setting of bony glenoid deficiency.



Fig. 52.1 An anteroposterior radiograph of capsulorrhaphy arthropathy after anterior stabilization.

In contrast to most nonanatomic repairs, anatomic procedures including Bankart repair³² and capsulorrhaphy³³ have become the mainstay of treatment and have evolved significantly with use of arthroscopic techniques for primary and revision instability repairs; some persons argue that these procedures are the current standard of care. However, this paradigm shift from primarily open to arthroscopic techniques leaves unanswered questions about the long-term outcomes of open versus arthroscopic procedures, especially with regard to arthritis.

Long-term data indicate the presence of degenerative changes after the Putti-Platt procedure in 30% to 61% of patients at 9 to 26 years of follow-up.^{34–36} These results have led to the abandonment of this procedure; however, the Bristow-Latarjet reconstruction remains part of the current treatment algorithm even though it has a reported incidence of arthritis of as high as 49% at 14 years.^{16,37,38} It may be argued that several technical details in the techniques used in these procedures (specifically, complete subscapularis takedown) are not currently performed, which may greatly affect outcomes, although few studies refuting this high rate of arthrosis have been reported. However, a more recent study has demonstrated that, despite an overall incidence of arthrosis in 34% of patients, moderate or severe arthrosis was present in only 6% (which correlated with worse outcomes), whereas the patients with only mild arthrosis had no degradation of outcomes from joint damage after the Bristow-Latarjet procedure.³⁸ These studies have shown that the development of arthrosis may be more closely related to graft malposition than other factors previously thought to be responsible.^{38,39}

Chondrolysis

Chondrolysis of the glenohumeral joint is a devastating problem that has proven to be one of the most vexing diagnoses faced by shoulder surgeons during the past decade. It has many associated

(although not well proven) causes and is characterized by unexpected pain and loss of joint mobility for weeks to months, typically after arthroscopic shoulder surgery. Global dissolution of cartilage occurs on both the glenoid and humeral head, with the characteristic findings of joint space narrowing, periarticular bone edema, and subchondral cystic changes, although without osteophyte formation.^{40–43} Although this condition has been associated with the use of radiofrequency energy in the shoulder, low-grade infection, intra-articular injection of contrast medium, use of bioabsorbable suture anchors, and postoperative pain pump infusion, small case series and case reports have led to difficulty in extrapolating meaningful data with regard to the true etiology of this condition.^{41,44–46}

Recently, 375 patients who underwent an arthroscopic intra-articular shoulder procedure by an individual community orthopedic surgeon were evaluated specifically with regard to chondrolysis.⁴⁷ This cohort demonstrated a 13% incidence of chondrolysis, with all cases associated with an intra-articular postarthroscopic infusion of local anesthetic (bupivacaine or lidocaine); chondrolysis did not develop in any patients who underwent arthroscopy without a postoperative anesthetic infusion and is not present following an intra-articular steroid injection.⁴⁸ In addition, Bankart repair, arthroscopic débridement, suture anchor placement in the glenoid, and operative time were associated with this finding. This finding of the association between a postoperative anesthetic infusion and chondrolysis has been further supported by the results of a recent systematic review of 100 cases of chondrolysis in the literature, with this situation demonstrated in 59% of the cases reviewed.⁴⁹

In addition, it has been demonstrated that the development of chondrolysis after a postoperative anesthetic infusion may be a dose-related phenomenon, which has been demonstrated in several basic science studies^{50,51} and supported in a recent clinical cohort. When a high-flow pain pump was used postoperatively, chondrolysis developed in 16 out of 32 patients, whereas it developed in only 2 out of 12 patients with a low-flow pump containing bupivacaine, epinephrine, and infusate.⁵²

With regard to low-grade infection as a causative agent, in their systematic review of 100 cases, Scheffel et al.⁴⁹ reported that of 91 patients who underwent revision surgery, 41 had microbiologic cultures taken at the time of revision surgery. Growth was reported for only 3 of these 41 cases, with *Propionibacterium acnes* reported in all 3 cases and no finding of any other positive cultures mentioned.⁴⁹

The possibility that bioabsorbable suture anchors are an etiologic agent has also recently been analyzed; Dhawan et al. performed an evaluation of the 1,072,000 bioabsorbable anchors placed about the shoulder in 2008.⁵³ Of these cases, 10 suture anchor-related complications were reported to the US Food and Drug Administration, all of which were thought to be related to anchor malposition and typically demonstrated geographic rather than global cartilage loss. This finding was viewed as a best-case scenario, in that many cases of complications may not be reported; however, the authors still concluded that use of bioabsorbable anchors remains a safe practice when they are implanted properly based on the available literature.

Unfortunately, a reliable treatment algorithm or diagnostic modality to definitively identify patients with chondrolysis does not currently exist. Chondrolysis remains largely a diagnosis of exclusion, although surgeons need a high index of suspicion in the setting of failed previous surgery with early, progressive, and advanced arthritic changes of the glenohumeral joint. The outcomes of further definitive treatment of these patients have also been disappointing, and chondrolysis thus remains a major challenge for shoulder surgeons.

Rheumatoid Arthritis

Rheumatoid arthritis is an inflammatory arthritic condition that affects the synovial linings of both small and large joints. Although other inflammatory arthritides may affect the young athletic population (particularly psoriatic arthritis, spondyloarthropathy, and reactive arthritis), most is known about rheumatoid disease in this setting. The inflammatory nature of the disease leads to a disabling secondary erosive arthritis. It most commonly affects the small joints of the body but is found in the shoulder in 90% of patients with chronic rheumatoid disease.⁵⁴ Rheumatoid disease in the shoulder is progressive in nature, with early symptoms of pain, swelling, and decreasing shoulder motion. With disease progression, extra-articular structures become involved, including the subacromial bursa, acromioclavicular (AC) joint, and rotator cuff.⁵⁵ Radiographic progression begins with medial migration of the humeral head into the glenoid with characteristic central erosion of the articular surface (Fig. 52.2). Bone quality is typically osteopenic with periarticular erosions involving the superior and medial humeral head. Cystic formation may occur at the rotator cuff insertion, which, coupled with intrinsic degeneration of the rotator cuff, may lead to functional rotator cuff deficiency, static superior migration, and uneven joint erosion. Late sequelae of the rheumatoid shoulder leads to painful joint destruction, loss of



Fig. 52.2 An anteroposterior radiograph of central erosion seen in persons with rheumatoid arthritis.

bone stock (which has significant implications for reconstructive options), rotator cuff compromise, and poor overall function.^{55,56} Further evaluation, especially in the preoperative setting, of all patients with rheumatoid arthritis should also include flexion and extension lateral views of the cervical spine to check for cervical instability and further causes of rheumatologic problems that may affect the shoulder joint concomitantly.

Osteonecrosis

Osteonecrosis (also known as avascular necrosis [AVN]) is another form of progressive glenohumeral arthritis that may be encountered during evaluation of the shoulder of a young athlete with arthritic disease. Osteonecrosis is characterized by the development of avascular regions of periarticular bone, resulting in infarction, necrosis, and ultimately collapse of the bony architecture of the subchondral plate with resultant deformity and arthrosis as a result of incongruity. Early workup of this disease should include testing for sickle cell anemia, particularly in the African-American population. Also to be considered etiologically are protein C and S deficiency, factor V Leiden, and hyperlipidemia.

Although many causes of osteonecrosis have been described, including dysbarism, hemoglobinopathies, coagulopathies, Gaucher disease, and connective tissue disorders, steroid use and alcohol abuse predominate clinically.^{57,58} If radiographs reveal a crescent sign and collapse, the contralateral shoulder should be imaged to rule out asymptomatic bilateral disease, and use of the aggravating agent (if identified) should be stopped to prevent further progression of the disease. Collapse of the articular surface is secondary to fractures in weak subchondral bone, and pain is often the most common complaint, specifically with difficulty sleeping and pain interfering with activities of daily living. Pain is usually present with flexed and abducted shoulder positions because of the incongruous central superior position of the humeral head contacting the glenoid in this position.

Classification of osteonecrosis of the humeral head has been described by Cruess (Table 52.1).⁵⁹ Osteonecrosis has six stages based on radiograph and magnetic resonance imaging (MRI); these stages are largely based on and parallel the original description in the femoral head by Arlet and Ficat.⁶⁰ Indeed, osteonecrosis of the humeral head is similar to that in the femoral head, first with edema visible only on MRI, followed by sclerosis and then development of the crescent sign, with further collapse of the

articular surface and finally degeneration on both sides of the joint (Fig. 52.3).

PATIENT EVALUATION

Presentation and History

Glenohumeral arthritis has no classic or pathognomonic history but typically presents as a progressive loss of shoulder rotary mobility coupled with pain. Some persons report noise, grinding, crepitus, and a feeling of a point of “clunking with release.” Another suggestive history includes instability without a history of dislocation, with the incongruent articular surfaces having a mechanical catch. In addition, patients may report morning stiffness that improves throughout the day, along with pain during sleeping.

Primary glenohumeral arthritis in the young population (i.e., younger than 50 years) is exceedingly rare and warrants an exhaustive history to identify the underlying primary diagnosis. History should include prior episodes of trauma, current and previous medications, family history, prior surgical procedures, and recreational and social factors. In addition, the patient’s desired activity level should be taken into consideration. We have found that the desired activity level is a critical aspect of the evaluation of these patients because outcomes from certain procedures can be perceived as “catastrophic” if patients are not counseled about the longevity and limitations of implants and their ability to meet patients’ expectations.

Physical Examination

Examination of the shoulder starts with evaluation of the cervical spine, with notation of deficits in range of motion (ROM) and reproduction of pain with provocative maneuvers, such as the Spurling test. Referred pain from the cervical spine is common; patients with cervical spine pathology are often more comfortable

TABLE 52.1 Stages of Humeral Head Osteonecrosis (Cruess Classification)

Stage	Description
I	Normal radiograph; changes on magnetic resonance imaging
II	Sclerosis (wedged, mottled), osteopenia
III	Crescent sign indicating a subchondral fracture
IV	Flattening and collapse
V	Degenerative changes extend to glenoid

From Cruess RL. Experience with steroid-induced avascular necrosis of the shoulder and etiologic considerations regarding osteonecrosis of the hip. *Clin Orthop Relat Res.* 1978;130:86–93.



Fig. 52.3 Humeral head collapse in a patient with stage IV osteonecrosis.

with their hand draped over their head. Although referred pain and findings of cervical spine pathology do not exclude shoulder pain, patients suspected of having cervical pathology should be referred for proper care and the possibility excluded from the differential diagnosis.

The shoulder examination begins with visual inspection for symmetry, muscle atrophy, and previous surgical incisions. It is critical that the entire shoulder be visualized, including the entirety of the scapula, to ensure that the patient is examined for subtle but significant pathology, including scapular winging. Next, palpation of the posterior capsule, AC joint, greater tuberosity, and biceps groove helps to delineate and isolate glenohumeral pathology. Further evaluation includes both passive and active mobility with careful delineation of the painful arc of motion, while looking for the typical finding of an equal limitation of active and passive ROM in all planes. Motion is evaluated in forward flexion, abduction (controlling for scapulothoracic motion), external rotation at zero and 90 degrees of abduction, and internal rotation with regard to the vertebral level reached by the hand behind the back. Findings should be compared with the contralateral (presumably unaffected) side. Decreased rotational movements and mid ROM pain should alert the examiner to the presence of an arthritic process, whereas terminal ROM pain is more typically indicative of impingement, osteophytes, and capsular contraction.

Strength testing and impingement signs aid in the evaluation of rotator cuff pathology. Each tendon of the rotator cuff must be evaluated independently. The supraspinatus is evaluated using the champagne toast test,⁶¹ whereas infraspinatus testing is accomplished via external rotation strength with the arm adducted at the side.⁶² Lag signs must be elicited if they are present, and the Hornblower sign (external rotation with the shoulder abducted 90 degrees) tests the integrity of the teres minor.⁶³ In this patient population, we have found that it is critical to evaluate the integrity of the subscapularis, especially in the setting of capsulorhaphy arthropathy after an open anterior instability procedure, where we have noticed a high rate of this finding.^{64–66} Subscapularis tear is typically characterized by asymmetric hyperexternal rotation with the arm in adduction with a positive belly press (a test for the upper subscapularis) and lift-off (a test for the lower subscapularis).⁶⁷ The implications of these findings are critical because they may play a significant role in surgical decision-making; for example, consideration of procedures such as a concomitant pectoralis major transfer in the setting of subscapularis insufficiency with arthroplasty may be necessary.

Evaluation of Prior Operative Notes and Arthroscopic Images

A complete evaluation of the dates of individual surgeries, prior imaging, and arthroscopic images should also be undertaken for each patient. Such an evaluation can help differentiate between existing pathology and a new diagnosis. Particular attention to results after prior procedures is imperative to understand if recurrent pathology is present or if further problems have developed. Arthroscopic images help provide a firsthand view of the cartilage surfaces and the previously treated pathology. Previous MRI scans and radiographs can help put together a picture to

further understand previous and new pathology. A review of the operative report of previous surgeries can help the surgeon understand the exact surgical procedures and difficulties affecting the surgical outcomes. Patients often understand that they have a problem in their shoulder but do not effectively communicate exact pathologies that may help guide future considerations.

Careful examination of the detailed operative note should include notation of implants, intraoperative findings, and difficulties with the previous procedures. Prior implants could limit the ability to access the canal if, for example, a biotenodesis screw has been previously placed for a biceps tenodesis. Placement of metal suture anchors into the glenoid and greater tuberosity also could be limiting. These anchors may scratch sensitive metal and polyethylene components that would lead to early failure of further implants, without removal of the previous anchors. In addition, certain glenoid anchors have been thought to cause cyst formation, which may make it difficult to place a stable glenoid component on the subchondral surface.

Imaging

Evaluation of all types of arthritis starts with plain film radiographs. X-ray views should include at a minimum a Grashey anteroposterior view, an axillary lateral view at 45 degrees of abduction, and a scapular Y view (Fig. 52.4). These three views are able to document the position of the humeral head in relation to the glenoid, the presence of osteophytes, bone quality, the glenohumeral joint space, and glenoid bone loss. They should be evaluated for humeral head posterior subluxation, which may be the earliest sign of arthritis found on radiographs.¹¹

Further evaluation includes advanced modalities such as computed tomography (CT) and MRI. CT arthrography is extremely helpful in evaluating joints, especially in the setting of prior hardware placement. Glenoid morphology, bone stock and quality, and the status of the cartilage and rotator cuff can be well evaluated with a CT arthrography scan. A glenoid version can be further assessed by the techniques described by Friedman and colleagues (Fig. 52.5).⁶⁸ Walch et al.^{11,69} further described the bony anatomy of posterior glenoid wear in the anteroposterior plane, specifically evaluating for biconcavity, which is best evaluated by CT (Fig. 52.6). MRI is especially helpful in evaluating changes to the subchondral bone and associated soft tissue comorbidities.^{70,71} However, the sensitivity and specificity of evaluating chondral lesions is generally regarded as poor; up to 45% of grade IV chondral lesions may be missed.^{72,73} Newer evaluations with 1.5T and 3.0T MRIs have been found to detect glenoid and humeral lesions at 84% and 78%, respectively, with lower-grade lesions often missed.⁷⁴

TREATMENT OPTIONS

Treatment goals for the younger athletes with glenohumeral arthritis are resolution of symptoms and restoration of function and mobility. However, patient expectations must be matched to the durability and longevity of the specific treatment chosen. In fact, in 2008 McCarty et al.⁷⁵ demonstrated that the most common reason this population pursued treatment was to attempt to return to their previous level of sporting activity. Although

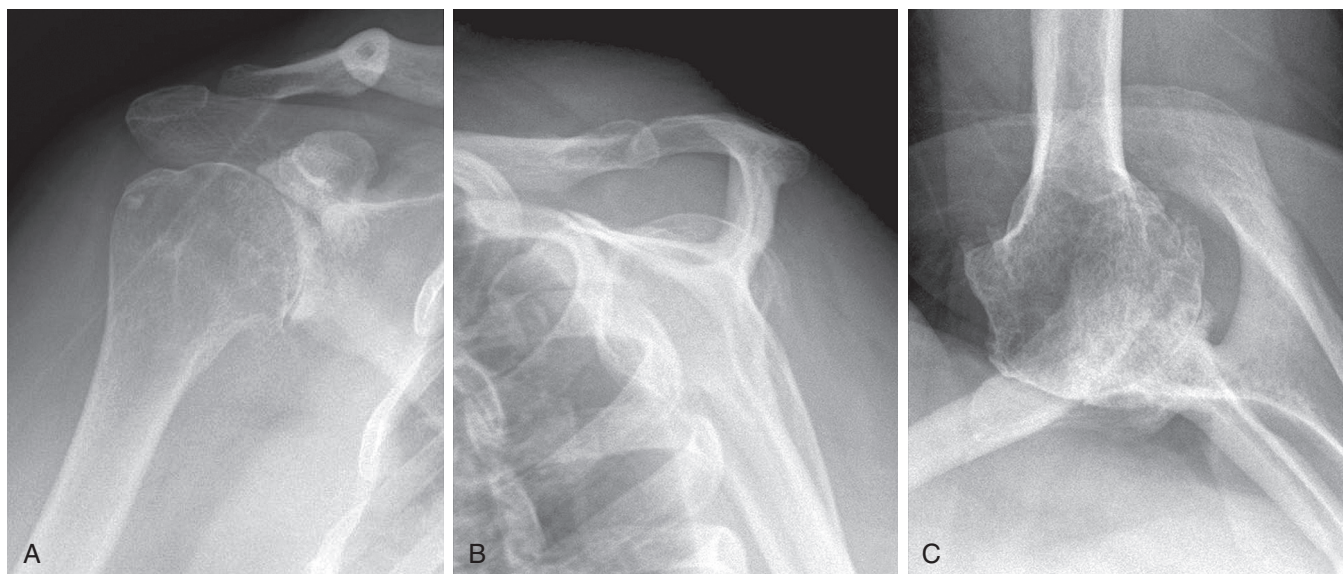


Fig. 52.4 A standard radiograph series. (A) Anteroposterior. (B) Scapular Y. (C) Axillary lateral.

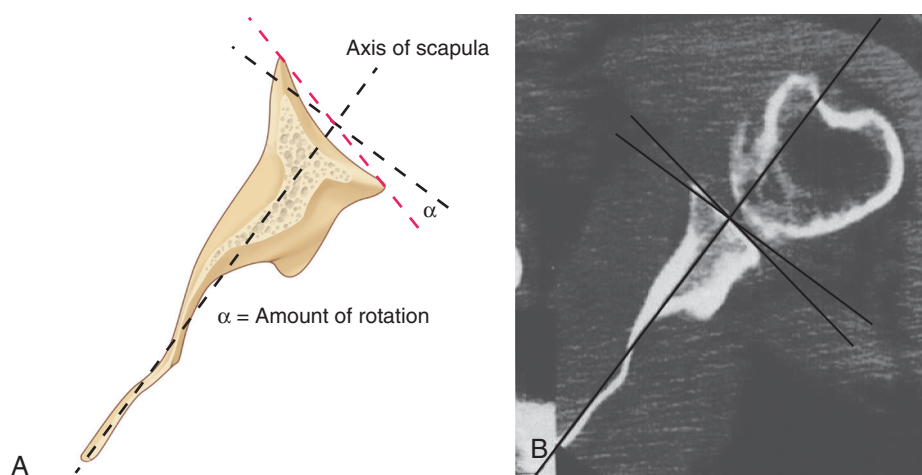


Fig. 52.5 The method of measuring the glenoid version as described by Friedman et al. (From Friedman RJ, Hawthorne KB, Genez BM. The use of computed tomography in the measurement of glenoid version. *J Bone Joint Surg Am.* 1992;74:1032–1037.)

older patients have reliable outcomes with shoulder replacement, young glenohumeral arthritis patients have worse arthroplasty outcomes,⁷⁶ and their treating surgeons have difficult decisions to make with regards to implant survivability. The Neer-type total shoulder arthroplasty (TSA) implants have demonstrated an 84% survival rate at 15 years,⁷⁷ although revision surgery in the area of the shoulder is extremely complex and few implants have been able to compensate for the glenoid bone loss frequently encountered in revision settings. In young patients, nonoperative treatments should be thoroughly pursued and exhausted before surgical interventions are contemplated. The mainstay of successful outcomes, whether surgical or nonsurgical, is related to patient education that clearly outlines the natural history and functional process that is present, along with the provision of acute and chronic pain management.

Nonoperative Treatment

The initial treatment of symptomatic glenohumeral osteoarthritis involves recommendation of activity modification, supervised physical therapy, and a trial of oral nonsteroidal antiinflammatory medications. Physical therapy should include a daily regimen of exercise with a focus on strengthening periscapular, deltoid, and rotator cuff musculature. Stretching the joint with manual manipulation helps to improve ROM, which counteracts restrictions caused by cartilage damage and resultant capsular contracture. In addition to these modalities, image-guided or blind injection into the glenohumeral joint with a mixture of corticosteroid and lidocaine is an excellent diagnostic and therapeutic tool. However, such an injection may not provide long-term relief for athletic patients engaging in high-demand sports who

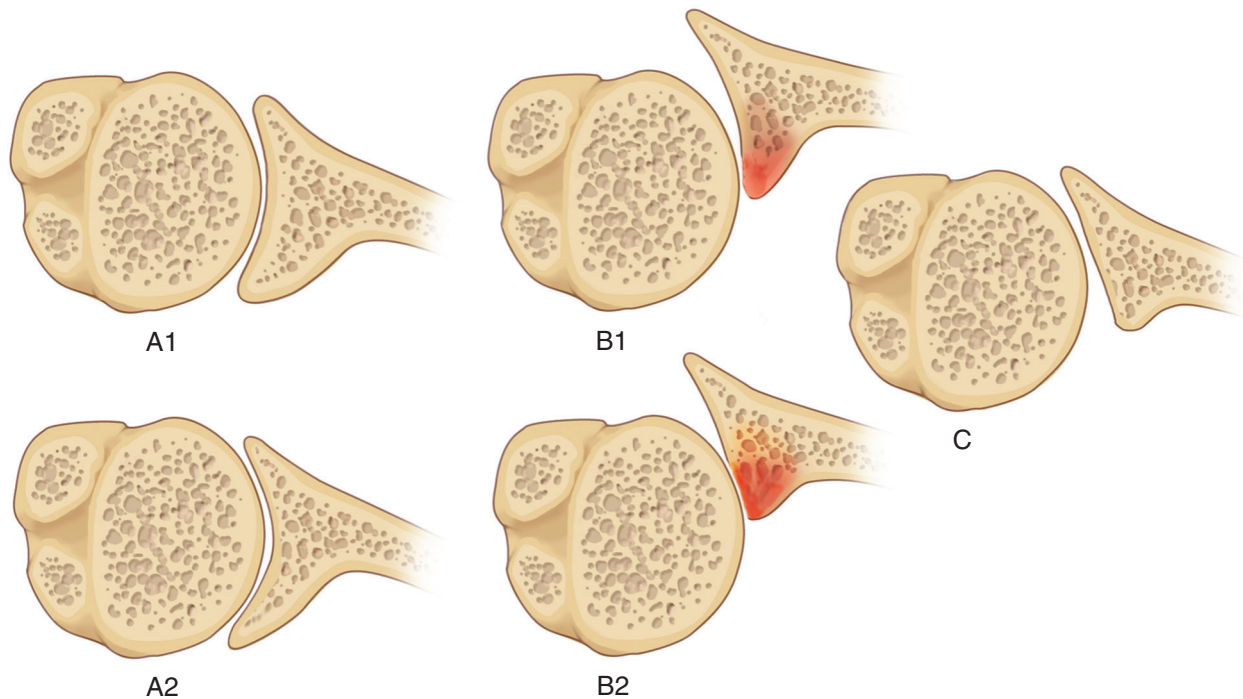


Fig. 52.6 The classification of glenohumeral morphology as described by Walch et al. (From Walch G, Badet R, Boulahia A, et al. Morphologic study of the glenoid in primary glenohumeral osteoarthritis. *J Arthroplasty*. 1999;14[6]:756–760.)

continue to participate in athletic endeavors, because their symptoms often return with continued activity.⁷⁸ In addition, off-label use of viscosupplementation with hyaluronic acid injections has been shown to provide partial symptom relief⁷⁹ but has not been shown to provide statistically improved outcomes over saline injections.⁸⁰ The relief experienced often involves improvement in overall pain and has been effective in allowing patients to sleep through the night with minimal shoulder pain after treatment.

Operative Treatment

Joint-Sparing Techniques

Arthroscopic débridement. The diagnosis of glenohumeral arthritis is often made incidentally at the time of arthroscopic intervention aimed at other pathology, unless radiographic changes or underlying conditions are present.^{81–83} However, arthroscopy provides a minimally invasive means to evaluate and treat chondral lesions and concomitant pathology in the shoulder, allowing for early rehabilitation and return to activity in symptomatic patients who have not responded to nonoperative measures.⁸⁴ Unfortunately, arthroscopy in this setting remains a temporizing measure that does not alter the underlying disease process; however, it can provide pain relief and functional improvement for a considerable amount of time.^{73,85–87}

Several authors have reported successful results of arthroscopic débridement for glenohumeral arthritis and isolated cartilage lesions in patients younger than 55 years, and several prominent themes have emerged from this work.^{73,86,88,89} Results have been reliable and have correlated with the extent of disease in those younger than 50 years.⁹⁰ Risk factors for failure are chondral

lesions larger than 2 cm², grade IV bipolar lesions, residual joint space less than 2 mm, the presence of large osteophytes, and Walch B2, C glenoids.⁹⁰ Weinstein et al.⁸⁶ reported that pain relief and maintenance of function was sustained for 76% of patients at 34-month follow-up, and it was sustained in 88% of patients at 2-year follow-up in the series by Cameron et al.⁷³ Conversion to arthroplasty was required in 22% of patients at an average of 10.1 months after débridement in one series⁸⁹ and in 15% of patients at 20 months in another series.⁸⁸

Several concomitant procedures were performed in the aforementioned series, including bicep tenotomy/tenodesis, distal clavicle resection, subacromial decompression, microfracture, loose body/osteophyte removal, and capsular release. We have found that capsular release is extremely useful, and it has been recommended that capsular release be included in arthroscopic treatment if rotation is limited by greater than 15 degrees in any plane of motion.⁷³ Significant attention must be given to these aforementioned concomitant pain generators in the setting of arthritic disease, and furthermore, failure to address these pathologies has been demonstrated to produce worse postoperative results.^{91–95}

Microfracture. In addition to débridement, it has been recommended that full-thickness cartilage lesions be managed with use of microfracture, similar to that described in the knee (Fig. 52.7).⁹⁶ In 2009 Millett et al.⁹⁷ reported on 31 patients (average age: 43 years) with full-thickness cartilage lesions in the glenohumeral joint who underwent treatment using microfracture. At a mean follow-up of 47 months, the patients demonstrated significant improvement in the visual analog scale score, American Shoulder and Elbow Surgeons (ASES) score, and painless use

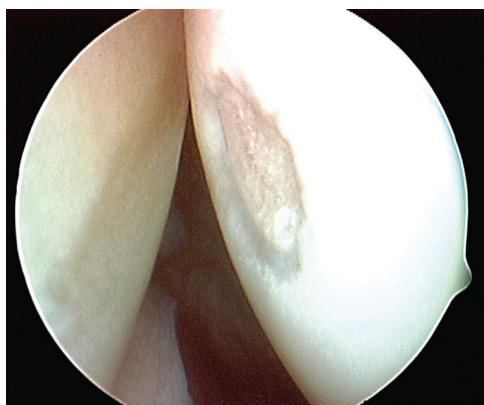


Fig. 52.7 An incidentally encountered isolated humeral head osteochondral lesion in a young patient undergoing arthroscopy for labral repair.

of the arm, as well as improvement in patients' ability to work, perform activities of daily living, and participate in sports activity, with failure in 19% of patients, who required additional procedures. These investigators demonstrated the greatest improvement in patients with smaller, isolated humeral head lesions and the worst results in patients with bipolar lesions.

More recently, Frank et al.⁹⁸ published their experience with a similar cohort of 15 shoulders in 14 patients (average age: 37 years) who underwent glenohumeral microfracture and demonstrated similar results. At an average follow-up of 28 months, these investigators showed significant improvements in the visual analog scale, simple shoulder test, and ASES scores, with 92% of patients stating that they would have the procedure again.⁹⁸ The procedure was considered a failure in three patients in this series (20%), who required further operative intervention during the study period.

Glenoidplasty and osteocapsular arthroplasty. In concordance with the concept of posterior subluxation and nonconcentric posterior glenoid wear, the technique of glenoidplasty was developed in an effort to restore glenoid morphology from a biconcavity to a single concave surface. Restoration of uniform concavity theoretically reduces posterior humeral head subluxation and increases the effective joint surface, thus reducing point loading while increasing stability. This technique is performed by using an arthroscopic burr where the anterior glenoid (or "high side") is resected until it is flush with the posterior joint.⁹⁹ Recommendations for this procedure are aimed at pain relief; however, it has been demonstrated that patients with pain in their mid-arc ROM do not fare as favorably after surgery as those who have end-ROM pain, pain at rest, and painless crepitus. Also associated with favorable results are the radiographic findings of a biconcave glenoid with posterior humeral subluxation and the presence of osteophytes and loose bodies. Contraindications to this procedure are glenoids without biconcavity and patients with severe pain in mid-arc motion, because these findings have been associated with severe arthritis and patients with these findings have not done well after glenoidplasty.

Osteocapsular arthroplasty involves removal of humeral osteophytes with release of capsular contractures. Complete capsular release from the rotator interval to the posterior-inferior capsular

recess should be performed to gain maximal mobility. In a cohort with an average age of 50 years who were treated with this procedure, Kelly et al.¹⁰⁰ demonstrated that more than 85% of patients reported improvement in both pain and ROM at an average 3-year follow-up. Elser et al.¹⁰¹ have briefly presented their data on 27 young patients who underwent a similar procedure that they have termed *comprehensive arthroscopic management*, which includes débridement, capsular release, and humeral osteoplasty. One-year follow-up demonstrated a patient satisfaction rate of 8.5/10 with the procedure, with only one patient progressing to arthroplasty in that period.¹⁰¹ In another series Millet et al. reported 76% 5-year survival of a comprehensive arthroscopic management (CAM) procedure, including débridement of degenerative labrum, biceps tenodesis, burring of inferior humeral head osteophytes, and occasional axillary nerve decompression. ASES and Single Assessment Numerical Evaluation (SANE) scores indicating improved pain score and function with work, activities of daily living, recreation, sleep, and use of the arms all significantly improved.

Arthroscopy for rheumatoid arthritis and osteonecrosis. Arthroscopy has also proven useful in the early treatment of both rheumatoid arthritis and osteonecrosis. The hypertrophic synovium of rheumatoid disease leads to bony destruction, and thus a synovectomy may help slow the disease progression. Synovectomy results in increased motion and decreased pain in 80% of patients; however, this procedure is only useful early in the disease process, and when radiographic signs of disease are present, synovectomy is no longer a viable option.^{102–105} Concomitant procedures should be performed as indicated, including subacromial decompression and rotator cuff repair, and the synovium may repopulate, requiring a repeat synovectomy.

Patients with osteonecrosis have also benefited from arthroscopy and core decompression early in their disease process, and results have been shown to correlate to the severity of humeral head involvement.^{106,107} This technique was first described in 1993 by Mont et al.¹⁰⁸ and has been shown to be successful in 94% and 88% of patients with stage I and II disease, respectively. These results decline to 70% and 14% success in patients with stage III and IV disease, respectively. Core decompression may delay the time until arthroplasty in persons with stage III disease, although indications are controversial and this procedure is contraindicated in persons with stage IV and V disease.

Joint Resurfacing Techniques

Humeral head resurfacing, biologic resurfacing, and osteochondral grafting. Secondary treatment options include humeral head resurfacing, biologic resurfacing of the glenoid, and osteochondral allograft or autograft of the humeral head or glenoid. These techniques are used along the spectrum depending on the amount and chronicity of injury to the articular surfaces. Also involved is the level of instability present as a result of underlying pathology. Osteochondral allografts have been suggested as a means for reconstructing defects in both the glenoid and humeral head in cases of instability of the shoulder, particularly in the setting of large Hill-Sachs lesions (i.e., >45% of articular surface). In the chronic setting, or when larger humeral defects are present, a structural allograft can be used as a biologic

means of extending the range of contact with the glenoid surface in external rotation.¹⁰⁹ Algorithms for treating humeral head defects have been well described.¹¹⁰ For older patients with more than 45% of the humeral head affected, a prosthetic replacement is recommended. For younger patients with greater than a 45% structural defect, an osteochondral allograft should be considered (Fig. 52.8). Evaluation of outcomes has largely relied on case reports demonstrating success at 2-year follow-up.^{111,112} Given the rarity of these cases and limited reports, each patient should be approached individually and treatment should be determined on a case-by-case basis. Allograft reconstruction techniques isolated to the glenoid have been shown to be beneficial at 5- to 7-year follow-up.

Humeral head resurfacing has been reviewed as a further consideration for patients who are young and active. Since its inception in 1979, this treatment has undergone numerous changes, including the addition of hydroxyapatite coating in 1993. Current techniques including the “ream and run,” designed to promote articular congruity, were first studied by Matsen et al.¹¹³ in a canine model and demonstrated fibrocartilage formation on the glenoid at 6 months. This model was then transitioned into a human population, and results were reported in 2007.¹¹⁴ They demonstrated equivalent subjective shoulder test values at 3-year follow-up when compared with TSA. An important note regarding this technique is that the glenoid should be normalized in morphology, and if posterior subluxation is present, the physician should not perform a posterior release and the inferior glenohumeral ligament should be left intact.

Bailie et al.¹¹⁵ in 2008 and Tibone et al.¹¹⁶ in 2015 again looked at results after isolated humeral resurfacing and found that patients uniformly had improved outcome scores at 3-year follow-up. Bailie et al.¹¹⁵ looked at a subset of patients younger than 50 years, which suggests that resurfacing is an acceptable treatment option for patients younger than 50 years who would like to maintain an active lifestyle. Levy and Copeland¹¹⁷ also looked at a subset of patients younger than 50 years and concluded at an average follow-up of 8.2 years that ROM and Constant scores were equal to the outcomes of TSA. These studies provide

validated outcome measures that give the treating surgeon an alternative to traditional shoulder replacement in the younger population without concern for glenoid component loosening.

Biologic resurfacing that focuses on glenoid “coverage,” combined with a humeral-sided hemiarthroplasty or resurfacing, has also been proposed as a solution to glenoid longevity and ultimate implant survival in the young, active population. Several glenoid resurfacing techniques have been proposed and studied, including use of a fascia lata autograft, the anterior shoulder capsule, an Achilles allograft, a meniscal allograft, and xenograft tissue (Fig. 52.9).^{118–121} However, the results of these techniques have been mixed. Although the techniques engendered much early enthusiasm because of the initial reports by Nowinski¹²⁰ demonstrating that 81% of patients had good or excellent results, with 21 of 26 patients returning to predisease activities (including heavy lifting and manual labor), more recent reports have demonstrated conflicting results. A longer-term follow-up study by Krishnan et al.¹²² showed excellent results in 18 of 36 patients (50%), satisfactory results in 13 of 36 patients (36%), and unsatisfactory

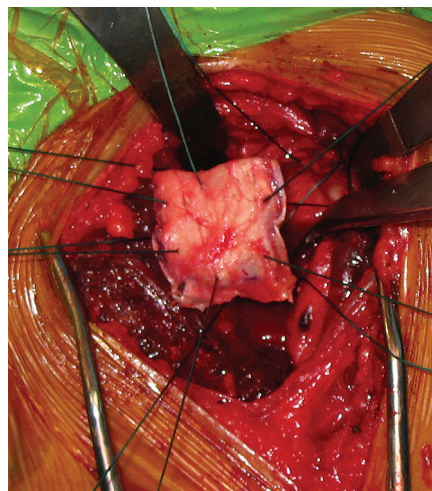


Fig. 52.9 Fascia lata autograft biologic glenoid resurfacing.

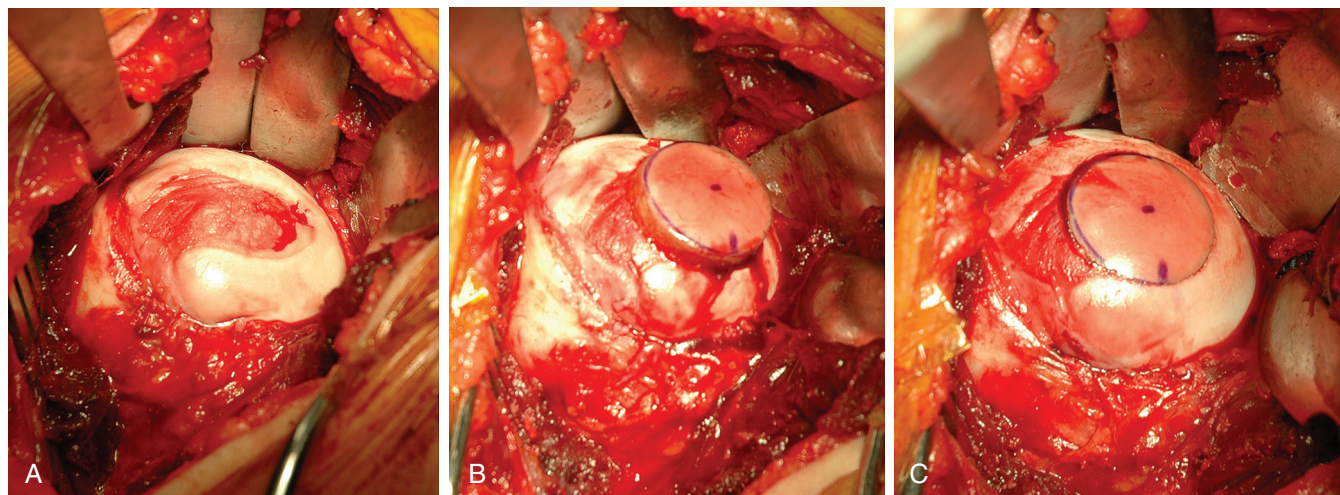


Fig. 52.8 (A) A large osteochondral lesion of the humeral head in a young patient. (B) Matched osteochondral allograft preparation. (C) Completed allograft osteochondral resurfacing.

results in 5 of 36 patients (14%) at 2- to 15-year follow-up with use of an anterior capsule, fascia lata, or Achilles allograft. These investigators concluded that use of capsular tissue was an independent predictor of failure, and thus they recommend use of an Achilles allograft in this setting. Similarly, Wirth¹²³ reported on 27 patients who underwent resurfacing with a lateral meniscal allograft and demonstrated significant improvements in pain relief and function at an average of 3 years of follow-up, with a 70% survivorship at 8 years.¹²⁴ However, this group demonstrated uniform progression of joint space narrowing, which calls into concern the long-term functional outcome in this cohort. These mostly positive results must be contrasted with results reported by Elhassan et al.,¹²⁵ Romeo et al.,^{126,127} and Gobeze et al.¹²⁸ regarding reoperation rates. Ten of 13, 12 of 21 patients, and 7 of 16 patients, respectively, required revision TSA. These studies give validated, yet conflicting results as a surgical option that is attractive because it allows younger, more active patients to return to their previous level of activity, including weightlifting and manual labor, without concern for eccentric glenoid wear, although the long-term results may be unpredictable.

Glenohumeral arthroplasty. Shoulder arthroplasty pioneered by Neer in the 1950s, with a persistent evolution of prosthetic design and surgical technique, has continued to improve function in patients undergoing shoulder arthroplasty. Patients with primary osteoarthritis, posttraumatic arthritis, inflammatory arthritis, osteonecrosis, capsulorrhaphy, and dislocation arthropathy can be effectively treated with shoulder arthroplasty. Pain that has failed to respond to other interventions is the most common indication for arthroplasty; however, patients often obtain statistical improvement with regard to ROM, strength, and function as well. The status of the soft tissues, bony structures, and underlying etiology must be taken into account when counseling patients about expected results and when planning the details of operative intervention.

Primary osteoarthritis of the glenohumeral joint that is unresponsive to nonoperative or joint-preserving techniques is a well-supported indication for shoulder arthroplasty (Fig. 52.10). Godeneche et al.¹²⁹ reported excellent results in a series of 268 patients. Objective outcomes improved based on Constant scores, with 77% good or excellent results, motion gains were seen with 50-degree improvements in elevation, and subjective scores demonstrated 94% satisfaction rates.¹²⁹ In a large multicenter study of more than 600 patients, Edwards et al.³ showed clinically and statistically significant improvements for all variables in patients undergoing total and hemiarthroplasty of the shoulder, including Constant scores, pain, ROM, activity, and strength. Khan et al., in a 10-year prospective evaluation of patients who underwent TSA, found a mean improvement of 41 points in Constant scores and a 65-degree improvement in forward elevation for patients with primary osteoarthritis. Component survival was found to be 100% for the humeral component and 92% for the glenoid.¹³⁰ Raiss et al.¹³¹ evaluated younger patients with mean age of 55 years who underwent TSA for primary osteoarthritis and found a 40-point increase in Constant scores, 95% satisfaction rates, and 100% component survival at a minimum 5-year follow-up. These results are consistent with the results of other series on arthroplasty for



Fig. 52.10 An anteroposterior radiograph of total shoulder arthroplasty for primary osteoarthritis.

primary osteoarthritis and demonstrate predictable and reproducible improvements in all parameters evaluated, providing thorough evidence when counseling patients.

Both preoperative and intraoperative factors can affect results of shoulder arthroplasty for glenohumeral joint degeneration. Preoperative factors include underlying etiology, rotator cuff tears, fatty degeneration of rotator cuff musculature, and glenoid morphology. Fatty degeneration of the rotator cuff musculature may result in less favorable postoperative outcomes in patients with Goutallier grade 2 or higher infiltration in the infraspinatus and subscapularis, resulting in decreased forward elevation and strength.¹²⁹ Small tears of the supraspinatus have little effect on outcomes after arthroplasty.¹³² However, larger tears negatively affect active ROM and strength and have a more important effect on component biomechanics. Larger tears can make it difficult for the rotator cuff to maintain the center of rotation, which allows superior migration of the humeral head in relation to the glenoid. In the setting of TSA, this phenomenon can have profound effects on glenoid component survival, due to elevation of the center of rotation, with subsequent asymmetric contact mechanics of the glenoid component. This situation results in the “rocking horse” effect in the cephalocaudal plane from point loading of the superior aspect of the glenoid component, which has been demonstrated to be a significant factor in early glenoid component loosening and failure.¹³³

Posterior humeral subluxation and posterior glenoid wear are associated with primary glenohumeral osteoarthritis, resulting in altered glenoid morphology, and must be taken into account when considering arthroplasty for degeneration of the glenohumeral joint.^{8,134} Native glenoid morphology, as well as changes resulting from glenohumeral degeneration, may affect operative planning and decision-making regarding the type of implant. Hoenecke et al.¹³⁴ classified glenoid morphology, including neutral,

biconcave, worn, dysplastic, displaced, and angled. In this report, as well as in studies by Gerber et al.¹³⁵ and Walch et al.,¹³⁶ the glenoid version was not associated with humeral head subluxation. Humeral head posterior subluxation is associated with a biconcave glenoid, and authors have recommended caution when attempting to correct the glenoid version if the goal is to address humeral head subluxation, using soft tissue balancing through capsular releases for implant stability instead of relying on bony correction alone. Indeed, Levine et al.¹³⁷ demonstrated that outcomes of hemiarthroplasty were most closely correlated to the status and extent of posterior glenoid wear. An individual approach should be used for glenoid preparation, with cautious correction of glenoid deformity and version while maximizing preservation of glenoid bone stock and glenoid component fixation.

Both hemiarthroplasty and TSA provide pain relief, but TSA has been shown to be statistically superior. Early reports revealed improved results with TSA compared with hemiarthroplasty; however, sample sizes were too small to show statistical significance until a meta-analysis by Kirkley et al.¹³⁸ and a multicenter study by Edwards et al.³ confirmed superiority in nearly all outcome measures. TSA outperformed hemiarthroplasty with regard to Constant scores, pain, motion, strength, activity scores, active forward elevation, and external rotation. Humeral head replacement alone has been reported to have poorer results for pain and function, as well as increased revision rates, compared with TSA.¹³⁹ In addition, TSA was reported by Mather et al.¹⁴⁰ to be a cost-effective procedure, of greater utility to the patient, and of lower overall cost to the payer compared with humeral head replacement alone.

Although TSA and hemiarthroplasty alone have had predictable and satisfactory results in older patients, the results in younger patients have been less favorable in some reports. Sperling et al.⁷⁷ reviewed 78 Neer hemiarthroplasties and 36 Neer total shoulder replacements in patients younger than 55 years and found 36 unsatisfactory results (46%) in the hemiarthroplasty group and 14 unsatisfactory results (38%) in the total shoulder replacement group, underscoring the challenges of treating glenohumeral arthritis in the younger patient. However, Rais et al.¹³¹ reported a 95% satisfaction rate with no revisions in patients with an average age of 55 years and 5- to 9-year follow-up, and Burroughs et al.¹⁴¹ found an 86% satisfaction rate and a 9% revision rate at midterm follow-up in patients younger than 50 years. Saltzman and colleagues¹⁴² evaluated patients undergoing shoulder arthroplasty before and after age 50 years and found a much larger proportion of patients requiring arthroplasty for reasons other than primary osteoarthritis, with the most common reason being capsulorraphy arthropathy. It was also reported that the pathoanatomy associated with the younger patient group complicated the surgery, rehabilitation, and outcome after shoulder arthroplasty.¹⁴² Arthroplasty for chondrolysis has had similar results at 2-year outcomes with reoperation rates of 25%.¹⁴³ Considering that the majority of articles confirm the challenges of performing shoulder arthroplasty in younger patients, caution should be taken and alternative treatments should be considered and exhausted before making the recommendation to proceed with arthroplasty.

Given the complex nature of arthritis in the young patient and inconsistent results with TSA, hemiarthroplasty, and biologic resurfacing, and given the fact that the most common complication after TSA is failure of the glenoid component, the ream-and-run technique has been described as an alternative to potentially avoid some of these issues. This technique involves concentric reaming of the glenoid while leaving the labrum intact, allowing correction of glenoid deformity and denervating the glenoid articular surface. Preliminary results of this technique in all age groups demonstrated significant improvements in both pain and function.¹⁴⁴ Saltzman et al.¹⁴² also evaluated this technique in patients younger than 55 and found significant improvement in pain and function with a low revision rate and little progression of glenoid erosion. Clinton et al.¹⁴⁵ confirmed satisfactory results using this technique, reporting similar self-assessed outcomes with ream and run compared with TSA; however, recovery time was slightly longer in the ream and run group.

Arthroplasty for inflammatory arthritis. Shoulder arthroplasty has shown consistent improvements in pain, function, and quality of life in patients with inflammatory arthritis. Most studies have favored TSA over hemiarthroplasty for functional outcomes; however, this conclusion is controversial, with several studies reporting no difference or improved outcomes with hemiarthroplasty compared with TSA in this patient population.^{139,146–151} Arthroplasty can be considered for inflammatory arthritis of the glenohumeral joint, with the most favorable outcomes in patients with an intact or repairable rotator cuff and adequate glenoid bone stock.¹⁵² Late tears of the rotator cuff from the underlying disease process are not uncommon after a successful TSA, and therefore patients should be followed up yearly to monitor for and prevent early failure from rotator cuff insufficiency.

Arthroplasty for instability arthropathy. Approximately 9% of patients who have undergone shoulder arthroplasty have a history of shoulder instability.¹⁵³ Marx et al.¹⁵⁴ demonstrated that the likelihood of experiencing glenohumeral arthritis is 10 times higher in patients with a history of shoulder dislocation and that, after surgical treatment of the instability, the probability increased 20-fold. Neer et al.¹⁵ first reported on the results of shoulder arthroplasty for osteoarthritis after instability surgery with satisfactory to excellent results. Shoulder arthroplasty for arthritis after instability has shown favorable results with high satisfaction rates and marked improvements in pain, mobility (specifically external rotation and abduction), and quality of life; however, complication rates, including revision rates, are clearly higher than with primary osteoarthritis.^{76,155,156} Complications such as continued instability, loosening, and decreased ROM are more common than in primary osteoarthritis, with some reports of up to 40% complication rates, and revision rates in the literature range from 20% to 40%.¹⁵³

Given the less consistent results in this patient population, arthroplasty must be approached with greater caution, taking into account the younger patient age and associated pathoanatomy in the setting of dislocation arthropathy. Prior operations can alter the native anatomy and should be addressed at the time of the arthroplasty; for example, nonanatomic stabilization procedures can cause internal rotation contractures and require that

an anterior release be performed to address the loss of external rotation. Prior procedures can also alter the native biomechanics of the glenohumeral joint and likely are the most significant factors responsible for less favorable results after shoulder arthroplasty for instability arthropathy compared with primary osteoarthritis.¹⁵³ Glenoid defects may require bone restoration, and soft tissue balancing is critical to prevent eccentric loading of the glenoid component, which may cause abrasive wear and early failure. Internal rotation contracture is the most common soft tissue abnormality found and may require extensive anterior capsular release. Subscapularis lengthening should be considered in more severe cases. However, subscapularis lengthening procedures can cause internal rotation weakness and may increase the likelihood of ultimate subscapularis failure. A preoperative CT scan is critical when a patient has evidence of glenoid defects, and the surgeon's index of suspicion must be high when evaluating patients with instability arthropathy. Determination of the extent and location of glenoid defects should be made preoperatively so appropriate plans can be made to address bony defects at the time of arthroplasty. Despite higher complication rates and increased surgical complexity, patients with instability and postcapsulorrhaphy arthropathy¹⁷³ can have acceptable outcomes after shoulder arthroplasty, with predictable improvements in pain and function.

An additional specific patient population that warrants separate consideration consists of patients undergoing TSA after the development of chondrolysis. There have been two case series addressing this issue in the literature, Levy et al.⁴¹ and Sperling et al.¹⁵⁷ Levy demonstrated good to excellent results in 10 of 11 patients, whereas Sperling experienced a 25% reoperation rate at 2 years. These investigators demonstrated an opportunity for improvements in both pain and function but cautioned that progressive glenoid lucencies may develop, thus highlighting the difficulty surrounding definitive care of these patients and the still unanswered questions with regard to optimal management of this complex problem.

Arthrodesis. Glenohumeral arthrodesis is considered an end-stage salvage procedure indicated in the setting of shoulder pain, weakness, and instability that is not suitable for soft tissue or prosthetic reconstruction.¹⁵⁸ Some situations may arise in the younger active population for which glenohumeral arthrodesis is an option, given the limitations with regard to the potential durability and longevity of arthroplasty solutions. Arthrodesis is a reliable procedure in which bony fusion is obtained in nearly all patients. The optimal position of arthrodesis is 20 degrees forward flexion, 20 degrees abduction, and 40 degrees internal rotation. Most patients report pain relief after this surgery, although few are completely pain free.^{159,160} The functional results of arthrodesis unfortunately lag behind improvements in pain. Few patients are able to return to any type of overhead work, although many patients find their fusion functionally beneficial from their preoperative state and can typically reach their mouths and hair, cross the midline for dressing, and reach their back pocket on the affected side.¹⁶¹ Obviously, given the mobility restrictions after this procedure, return to sporting activities is limited; however, in the salvage situation, it may be an option and can be later reversed to shoulder arthroplasty when implant

longevity becomes an option.¹⁶² This particular outcome has not been well studied in the younger active population.

DECISION-MAKING PRINCIPLES

In general, in our practice, young, active patients who are diagnosed with glenohumeral arthritis are individually evaluated, specifically with regard to determining the disease etiology, pathology severity, and their personal functional demands and goals. Determination of the etiologic cause of the disease is often critical because first line treatment in these patients is often directed at treatment of the underlying or inciting disease process (e.g., infection, inflammatory arthritis, or instability).

Initially we prescribe a formal course of nonoperative treatment, including supervised physical therapy, activity modification, and nonsteroidal antiinflammatory drugs, and injections as necessary. All concomitant pathology is evaluated, and treatment is directed at these entities, specifically focusing on identified concomitant bicep and AC disease. Our threshold for using image guidance for these injections is low, and we commonly use such guidance because the osseous anatomy of these patients is often distorted, and routine office-based injection techniques are technically more challenging and less successful. Initial treatment with an intra-articular corticosteroid injection is attempted, and based on the clinical response, these injections are continued sparingly as needed, or glenohumeral viscosupplementation is considered. When patients fail these modalities, they are considered candidates for operative intervention.

Operative decision-making depends on the pathology, disease severity, and functional demands of the patient. As previously mentioned, selective injections are used both as a diagnostic and therapeutic tool, especially when planning treatment of the AC joint and long head of the biceps tendon. In patients who have pain in these areas and respond positively to injection, distal clavicle resection and biceps treatment (tenodesis vs. tenotomy) is indicated, respectively, with subpectoral tenodesis preferred. We prefer this approach in younger patients who have bicep groove pain, because it has been our experience that tenotomy or tenodesis above the transverse ligament has led to unacceptable rates of residual pain, and this approach has been supported by a recent report in the literature.¹⁶³ An arthroscopic approach is typically entertained first, especially if the patient demonstrates a well-centered glenohumeral joint with a congruent humeral head and glenoid articulation and greater than 2 mm of joint space. If capsular contracture greater than 15 degrees is present, we perform capsular release in a controlled manner with a radio-frequency device under direct visualization. Cartilage lesions are initially considered for treatment with microfracture or débridement. This decision is made on the basis of a combination of factors, although data to support firm cutoffs based on the size of the lesion are currently lacking. However, we do not perform microfracture in the setting of diffuse, uncontained cartilage loss, bipolar or "kissing" lesions geographically, and untreated instability.

Failure of arthroscopic débridement and the aforementioned techniques, including 6 months of postoperative rehabilitation, may be an indication for consideration of osteoarticular allograft

filling or partial metallic resurfacing of residual humeral head cartilage defects with a Hemicap device (Hemicap, Arthrosurface Inc., Franklin, MA). Focal lesions of the glenoid are treated similarly with regard to débridement and microfracture; however, resurfacing options are limited.

Diffuse lesions or bipolar disease in the setting of advanced DJD, osteonecrosis, inflammatory arthritis, or capsulorrhaphy arthropathy are treated with humeral head resurfacing or hemiarthroplasty and individualized glenoid treatment. We have used the ream-and-run technique successfully, especially in the setting of an incongruous glenoid surface such as when a Bristow-Latarjet procedure that has failed. Decision-making regarding resurfacing versus stemmed humeral components is based on the need for glenoid exposure and visualization and the quality of the metaphyseal bone support (as in AVN), with stemmed components favored if a concern about fixation exists, especially in older persons.

Additional consideration is given to the status of the periarthicular soft tissue, particularly in the setting of failed open anterior capsulorrhaphy and subscapularis deficiency. These patients demonstrate hyperexternal rotation from loss of the static restraint to external rotation and a positive belly press/lift-off sign. We have treated these patients concomitantly with a split pectoralis major tendon transfer, based on the recent anatomic study by Fung et al.¹⁶⁴; in our experience it has functioned well in the setting of future arthroplasty.

We have used an arbitrary age of 40 years old for patients for whom we will typically perform a traditional TSA with a cemented glenoid component because of a perceived long-term survival benefit, extrapolated from older populations that have been studied, because data in this particular population are lacking. However, this decision is highly individualized, and when patients wish to continue with heavy-lifting activities or concerns of early glenoid loosening predominate, glenoid component implantation is avoided.

Authors' Preferred Technique

Glenohumeral Arthritis

In the setting of the athlete with glenohumeral arthritis, our preferred surgical technique greatly depends on the individual demands and severity of the disease. In the setting of a focal full-thickness cartilage defect, arthroscopic treatment is preferred. This treatment focuses on microfracture as appropriate, débridement, capsular release (if contracture is greater than 15 degrees), and treatment of concomitant acromioclavicular and bicep disease with a preference for biceps tenodesis if the patient has biceps groove pain. Arthroscopic treatment is also the initial surgical treatment of choice in the setting of diffuse disease, although only

when a round humeral head, greater than 2 mm of joint space, and well-centered glenohumeral articulation are present. Upon treatment failure of this approach or in the presence of diffuse cartilage disease, our preference has been humeral head resurfacing with individualized glenoid treatment, using either a "ream-and-run" technique or formal pegged glenoid component implantation. If there is any question of the support for a resurfacing component, such as in the setting of avascular necrosis, a standard stemmed arthroplasty is performed. Figs. 52.11 and 52.12 provide a more detailed outline of our preferred surgical treatment.

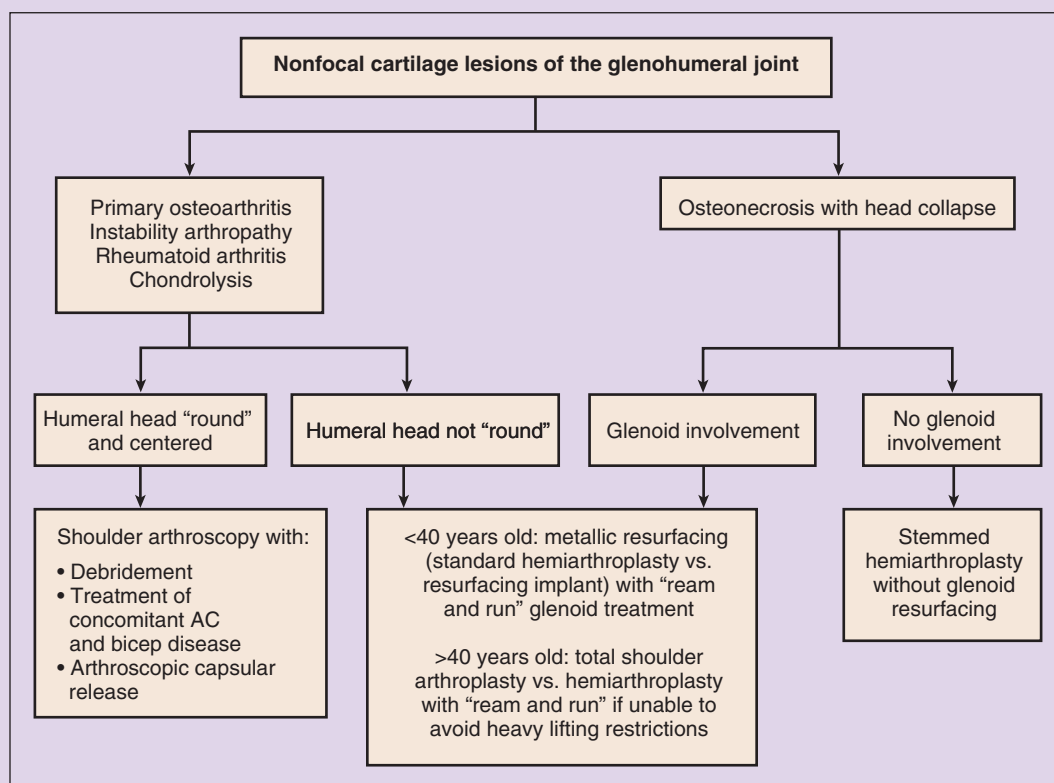


Fig. 52.11 The authors' preferred treatment of nonfocal cartilage lesions of the glenohumeral joint. AC, Acromioclavicular.

Authors' Preferred Technique

Glenohumeral Arthritis—cont'd

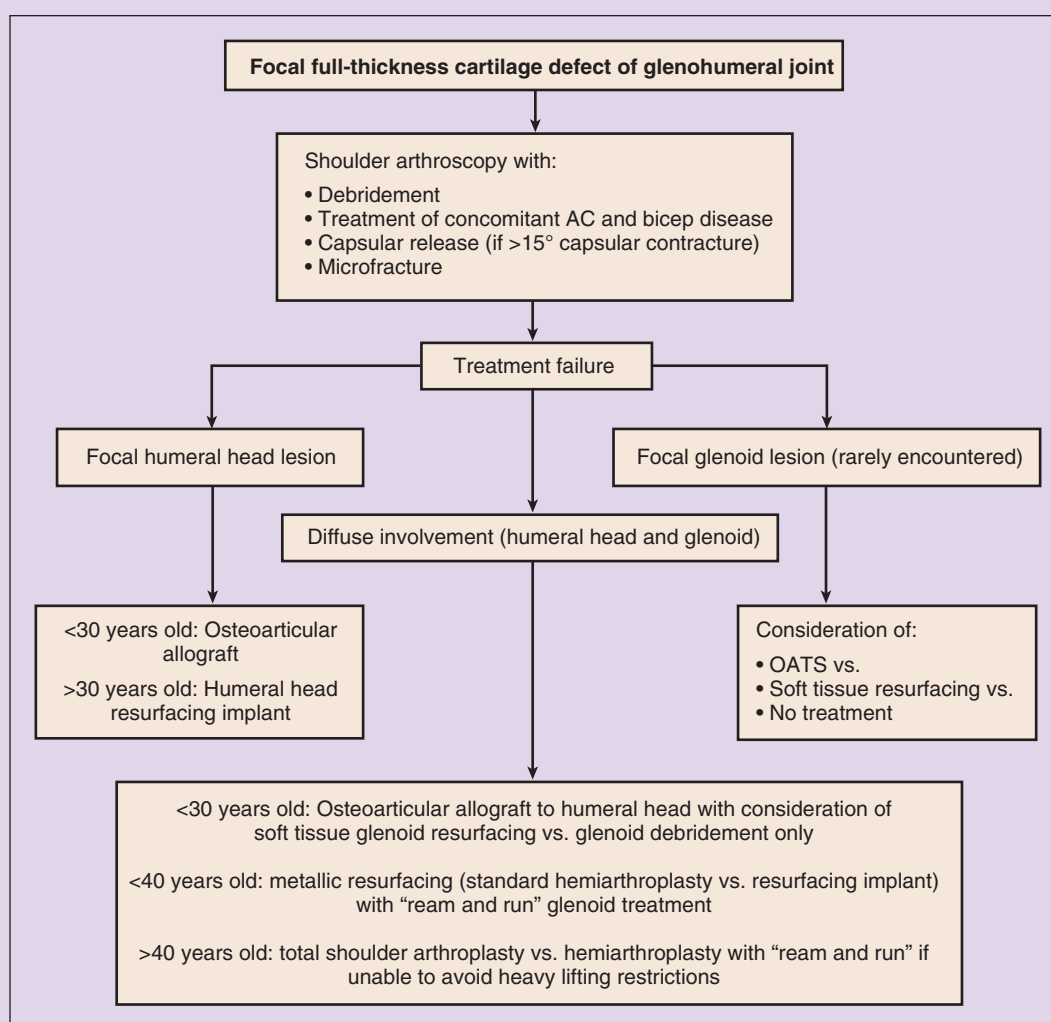


Fig. 52.12 The authors' preferred treatment of focal full-thickness cartilage lesions of the glenohumeral joint. AC, Acromioclavicular; OATS, osteochondral autograft transfer system.

POSTOPERATIVE MANAGEMENT

Postoperative Rehabilitation

Rehabilitation after arthroscopic intervention of the arthritic shoulder in the athlete is aimed at restoration and maintenance of functional ROM. Within 3 to 5 days, a regimen involving active, active assisted, and gentle passive ROM exercises is begun. Initially the goal is control of pain and edema. After this goal is achieved, ROM is emphasized. Strengthening is initiated after acceptable motion is achieved. This timing will often vary based on the extent of the patient's disease and preoperative ROM. This regimen may be performed with the assistance of a physical therapist or with a home exercise program in the case of a patient who is motivated and well educated on the proper regimen.

The surgical procedure (arthroscopic or open), the type of prosthesis used, and associated procedures performed determine

the type of orthosis used and the length of time its use is required in the postoperative period. Table 52.2 provides common orthosis protocols.

All patients, irrespective of the type of procedure, begin elbow, wrist, and hand mobility exercises daily, beginning on the first postoperative day.

Rehabilitation After Arthroplasty

The initial 6 weeks of the postoperative period consist of pain and edema control and prevention of excessive stiffness, while protecting the subscapularis repair. Irrespective of exposure technique, via subscapularis tenotomy or osteotomy, a sling should generally be worn for the first 6 weeks, removing it only for light deskwork, sedentary periods, bathing, or hygiene. A daily home program should be provided by the patient's therapist and performed two to three times per day. External rotation past 20 degrees and active internal rotation should be avoided to protect

TABLE 52.2 Postoperative Orthosis Protocols/Duration of Treatment

Procedure Performed	Orthosis	Duration (Weeks)
Arthroscopic procedure	Simple sling	1–2 as needed
Hemiarthroplasty or TSA	Simple sling	6
Resurfacing arthroplasty	Simple sling	6
Osteochondral grafting or partial resurfacing	Simple sling	6
Arthroplasty plus posterior capsulorrhaphy	Neutral rotation sling	6
Reverse total shoulder arthroplasty	Sling and abduction pillow	6

the subscapularis for the first 6 weeks. After 6 weeks, sling use can be discontinued and the patient may begin active ROM with reintegration of the extremity into activities of daily living. During this point in the recovery, patients should not lift, push, or pull more than 1 lb until 10 weeks after surgery but then may progress to 2.5 lbs per week. Isometric strengthening exercises may begin in the second week, with the exception of internal rotation. Throughout the entire recovery period, patients should not progress to the next phase of their rehabilitation protocol until they are proficient with the previous phase. The goal is to not simply complete the protocol but to maximize the functional results of each phase, and therefore timing of advancement to the next phase will vary based on the individual patient and the speed of his or her gains.

After sling use is discontinued, strengthening exercises are introduced, while the patient and therapist concentrate on ROM until functional mobility is achieved. However, ROM expectations must be based on preoperative ranges and often are unlikely to be normal.

Patients may begin sport-specific training with gradual progression at 12 to 16 weeks depending on the nature of the pathology and the procedure performed.

Return to Sport/Work

Activities of daily living generally resume by 6 weeks after surgical intervention for glenohumeral arthritis. Patients should be instructed to resume their normal activities gradually. This reintegration of the operated extremity into routine tasks serves to increase ROM, strength, and stamina with minimal risks. However, routine daily tasks may need to be specifically defined so athletes do not advance their activities beyond the protocol prescribed.

All patients are encouraged to pursue their preoperative activities and sports; however, we recommend strongly that patients avoid contact sports after a TSA. Before patients begin return to play (RTP) without restriction, we recommend completion of the rehabilitation protocol, as well as gradual progression of sport-specific activities and exercises, which helps facilitate the transition to RTP to prevent setbacks or reinjury. In a series of 75 patients evaluating RTP after shoulder replacement, McCarty et al.⁷⁵ reported that patients had a mean time to partial return

to sport of 3.6 months and a full return at 5.8 months. Weightlifting for maintenance of muscle tone is encouraged, but powerlifting exercises are discouraged. Recommendations for alteration of the frequency, duration, or intensity of the patient's desired activity or sport should be provided based on the individual and the sport being considered.

RTP expectations should be individualized to the athlete, pathology, and nature of the procedure to maximize the longevity of the intervention and minimize further trauma. Studies show that participation in sports after shoulder arthroplasty is high, and often a person's ability to participate is improved after surgery. Jensen and Rockwood¹⁶⁵ reported that 96% of patients in a series of recreational golfers were able to return to sport after shoulder replacement surgery. In their series, McCarty et al.⁷⁵ reported that 64% of patients cited sports participation as a reason they chose to undergo shoulder replacement. Swimming, tennis, and golf were the three most common sports in which patients participated, and 71% demonstrated an improvement in their ability to participate in their sport, with 50% of patients increasing their frequency of participation. However, a lower rate of return was noted in softball, weightlifting, and bowling. No difference was found in return to sport between hemiarthroplasty versus TSA. No association was found between participation in sports and requirement for further surgery or failure of the components at average 3.7-year follow-up. These findings, consistent with other reports of athletic participation after shoulder arthroplasty, provide practitioners with evidence-based support to aid in counseling shoulder arthroplasty candidates regarding postoperative participation in sport.¹⁶⁶

In addition, optimization of the kinetic chain should be encouraged through thorough completion of the rehabilitation protocol, as well as a maintenance program of home or gym exercises. Patients can continue to remain active after shoulder reconstruction; many of our own patients, as well as several studies, have demonstrated that a return to golf, tennis, snow skiing, mountain climbing, jogging, cycling, and many other physically demanding sports is possible.^{75,165,166} Patients have been able to return to work with hemiarthroplasty 69% of the time, with moderate intensity return to work of 50%. Patients have a higher risk of failure with higher upper extremity demand in sports such as softball, baseball, and basketball.¹⁶⁷ In Gulotta et al., 97% of TSA were able to return to sport versus 65% for HA at 2 years postoperative, with an average RTS at 5 months postoperative. These included swimming, baseball, softball, and basketball.¹⁶⁸

Complications

Complications in the treatment of glenohumeral arthritis depend on use of arthroscopic versus open techniques. Arthroscopic treatment has few complications, but complication rates of open procedures have been reported to be as high as 20% to 40%. Although many different problems may arise, they can often be avoided and solved by meticulous preoperative planning, careful surgical technique, and close attention to postoperative care. Tables 52.3 and 52.4 describe common complications and treatments for those complications in this setting.

TABLE 52.3 Common Intraoperative Complications and Treatment

Location	Complication	Treatment
Humerus	Diaphyseal fracture	Long-stemmed prosthesis Allograft struts and cables
Humerus	Greater tuberosity fracture	Suture fixation Postoperative modifications
Glenoid	Glenoid fracture	Rim—no treatment necessary Body—bone grafting/staged reconstruction
Rotator cuff	Supraspinatus/infraspinatus rupture	Repair may require reverse prosthesis if irreparable
Neurovascular injury	Axillary and musculocutaneous	Avoid overzealous retraction—observation
Vascular injury		Emergent vascular consultation for repair

TABLE 52.4 Common Postoperative Complications and Treatment

Location/ Diagnosis	Complication	Treatment
Wound	Hematoma	Irrigation and débridement if >7 days
	Dehiscence	Local wound care
Humerus	Component loosening	Infection workup
	Periprosthetic fracture	A. Revision longer stem B. Loose stem revision longer stem with allograft Well-fixed stem ORIF with allograft C. Loose stem—revision long stem component Well-fixed stem—trial nonoperative
Infection	Early, <6 weeks	Irrigation and débridement 6 weeks of IV antibiotics Attempt at component retention
	Late, >6 weeks	Staged procedure Removal of components 6 weeks of IV antibiotics
Instability	Anterior	Pectoralis major transfer Achilles tendon allograft Revision to reverse TSA
Instability	Posterior	Correction of humeral retroversion Increasing humerus head thickness
Glenoid	Loosening	Revision to pegged glenoid ± bone graft

FUTURE CONSIDERATIONS

Several novel ideas have been presented that may ultimately have applicability for the young, active patient with glenohumeral arthritis. The use of platelet-rich plasma is gaining in notoriety and popularity as several recent studies have begun to demonstrate the potential efficacy with regard to treatment of joint arthrosis.^{169,170} To our knowledge, the use of platelet-rich plasma has not yet been studied in the shoulder, although it may develop into a viable treatment option with further examination. Furthermore, advances in technique with regard to arthroscopic-based

resurfacing techniques have recently been presented¹⁷¹ and may represent an alternative or adjunct to current resurfacing options. Further studies into implant longevity based on design, specifically with regard to glenoid fixation, are also ongoing and may contribute to a significant long-term outcome benefit in this patient population. Another consideration is matrix-induced autologous chondrocyte implantation (MACI) for the shoulder. This has been provisionally studied in rabbits and has indicated results poorer to microfracture.¹⁷²

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

Edwards TB, et al. A comparison of hemiarthroplasty and total shoulder arthroplasty in the treatment of primary glenohumeral osteoarthritis: results of a multicenter study. *J Shoulder Elbow Surg.* 2003;12(3):207–213.

Level of Evidence:

III

Summary:

This study of 601 patients who underwent total shoulder arthroplasty (TSA) versus 89 patients who underwent hemiarthroplasty demonstrated the superiority of TSA with regard to pain, mobility, functional scores, and activity in persons with primary osteoarthritis.

Citation:

Samilson RL, Prieto V. Dislocation arthropathy of the shoulder. *J Bone Joint Surg Am.* 1983;65A(4):456–460.

Level of Evidence:

IV

Summary:

This article reports on a case series demonstrating high rates of arthrosis in unstable shoulders; the arthrosis was demonstrated to be worse in the setting of delay of diagnosis, nonoperative treatment, and posterior instability.

Citation:

Hovelius L, Saeboe M. Neer Award 2008: arthropathy after primary anterior shoulder dislocation—223 shoulders prospectively followed up for twenty-five years. *J Shoulder Elbow Surg.* 2009;18(3):339–347.

Level of Evidence:

IV

Summary:

This 25-year radiographic follow-up study of patients with primary anterior shoulder dislocation demonstrates that age at time of dislocation (25 years), recurrence of instability, participation in high-energy sports, and alcohol abuse are associated with greater risk of the development and severity of resultant arthropathy.

Citation:

Warner JP, et al. Glenohumeral arthritis in the young patient. *J Shoulder Elbow Surg.* 2011;20(6):s30–S40.

Level of Evidence:

IV

Summary:

This systematic review of the literature on the arthroscopic and open treatment options for early glenohumeral osteoarthritis reviews multiple treatment options and their outcomes.

Citation:

McCarty EC, et al. Sports participation after shoulder replacement surgery. *Am J Sports Med.* 2008;36(8):1577–1581.

Level of Evidence:

IV

Summary:

In this case series of 75 patients (average age, 65 years) who were followed up for 2 years, 64% reported “participation in sports” as a reason they decided to have surgery, with 71% reporting improvement in their ability to play their sport and 50% reporting increased frequency of participation. The most favorable sports for RTP were swimming, tennis, and golf, with softball demonstrating the lowest return (20%). The mean time to return to these activities was 3.6 months.

Citation:

Elser F, et al. Glenohumeral joint preservation: current options for managing articular cartilage lesions in young, active patients. *Arthroscopy.* 2010;26(5):685–696.

Level of Evidence:

V

Summary:

This article provides a comprehensive review of joint-preservation techniques for the shoulder, including a surgical decision-making guide for arthroscopy, microfracture, osteochondral autograft transfer system, autologous chondrocyte implantation, allograft, and biologic resurfacing.

Citation:

Levine WN, et al. Hemiarthroplasty for glenohumeral osteoarthritis: results correlated to degree of glenoid wear. *J Shoulder Elbow Surg.* 1997;6(5):449–454.

Level of Evidence:

IV

Summary:

This retrospective review of 31 patients who had undergone hemiarthroplasty for glenohumeral osteoarthritis demonstrates a satisfactory result in 74% of patients, which correlated directly with the amount of posterior glenoid wear. In patients with type I glenoids, 86% achieved satisfactory results, compared with 63% in patients with a type II glenoid, although similar pain relief was achieved in both groups, with the greatest deficits in forward elevation and external rotation.

Citation:

Saltzman MD, et al. Comparison of patients undergoing primary shoulder arthroplasty before and after the age of fifty. *J Bone Joint Surg Am.* 2013;95A(1):562–569.

Level of Evidence:

IV

Summary:

This instructional course lecture of the AAOS reviews possible complications of a TSA. In reviewing the complications, it also gives treatment options and intraoperative suggestions when dealing with particularly difficult situations that may prevent future complications.

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